



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 16 July 2026 01:01 AM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of the organization's workforce dataset, which contains records for 21 employees across 8 key attributes. The primary objective is to evaluate data quality, analyze workforce demographics and compensation structures, and identify potential risks and optimization opportunities.

The dataset exhibits high overall integrity with a Data Health Score of 92. However, minor data hygiene issues exist, including one duplicate row, one missing salary value, and one statistical outlier in experience.

The most striking analytical finding is the near-perfect correlation (ranging from 0.92 to 0.99) between age, experience, salary, and performance. This suggests an incredibly rigid, seniority-based operational and compensation structure. While highly predictable, this rigid alignment presents distinct retention and engagement risks for high-potential junior employees.

DATASET OVERVIEW

The dataset consists of 21 rows and 8 columns mapping key employee metrics. The schema is structured as follows:

EmployeeID: Unique identifier for each employee.

Department: Functional area (including IT, HR, Finance, and Sales).

Gender: Gender of the employee (e.g., Male, Female).

Age: Age of the employee.

Experience: Years of professional experience.

Salary: Annual compensation.

Performance: Performance score (scale assumed up to 100).

City: Geographical location of the employee (e.g., Bhubaneswar, Cuttack, Rourkela, Puri).

DATA QUALITY ANALYSIS

A detailed review of the dataset's hygiene was conducted, resulting in a solid Health Score of 92.

The following specific data anomalies were identified:

Missing Values: There is exactly 1 missing value in the `Salary` column. All other columns are 100% complete.

Duplicate Rows: The dataset contains 1 duplicate row that must be purged to prevent double-counting in payroll and headcount metrics.

Outliers: 1 potential outlier has been detected in the `Experience` column. This outlier requires manual verification to confirm if it represents a data entry error or a uniquely tenured employee.

STATISTICAL ANALYSIS

An analysis of the continuous variables revealed exceptionally strong, positive linear correlations across the board:

Age & Experience (0.99): As expected, age and years of experience move almost perfectly in tandem.

Experience & Salary (0.99) & Age & Salary (0.99): There is a near-perfect linear relationship between tenure/age and compensation. This indicates that salary increases are determined almost exclusively by seniority and age rather than merit.

Performance & Age (0.94) & Performance & Experience (0.92): Older, more experienced employees consistently receive higher performance ratings.

Salary & Performance (0.94): Higher salaries are strongly correlated with higher performance

evaluations.

BUSINESS INSIGHTS

Lock-Step Compensation Model: The 0.99 correlation between experience and salary confirms a traditional, seniority-based pay structure. Compensation adjustments are predictable but potentially unresponsive to market rate shifts or individual performance.

Potential Seniority Bias in Appraisals: The 0.92 correlation between experience and performance is unusually high. While experience typically enhances capability, a correlation this strong indicates that the performance appraisal framework may suffer from structural seniority bias, where tenure is conflated with actual output.

Geographic Footprint: Preview data indicates a localized workforce hub in Odisha (Bhubaneswar, Cuttack, Rourkela, Puri). This localized footprint can be leveraged for regional talent branding and localized office administration.

RISK ASSESSMENT

Retention Risk for High-Performing Junior Talent: If salary and performance scores are strictly tied to age and tenure, high-performing younger employees may feel undervalued. This can lead to the attrition of ambitious junior staff who feel their career progression is artificially capped by time-in-role requirements.

Operational Inefficiency & Complacency: A system that automatically rewards seniority with high performance ratings risks creating complacency among senior staff, while demoralizing junior employees who cannot achieve top-tier performance scores regardless of effort.

Data Integrity Risk: Leaving the duplicate row and missing salary value unresolved will distort average salary metrics, skew payroll budgeting, and compromise the accuracy of organizational reporting.

CONCLUSION

The workforce dataset is structurally sound and of high quality, as reflected by its Health Score of 92. Resolving the minor data anomalies (one duplicate, one missing value, and one outlier) will

elevate this dataset to perfect utility.

From a business perspective, the analysis reveals a highly traditional, seniority-driven organizational culture. While this provides stability and clear career pathways, it introduces significant retention risks for younger, high-performing staff. By modernizing the performance appraisal framework and introducing merit-based incentives, the company can maintain the stability of its senior workforce while energizing and retaining its junior talent.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
Version	2.0.0
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%
Duplicate Rows	1

DATASET HEALTH SCORE

Overall Health Score : 92%

MISSING VALUES

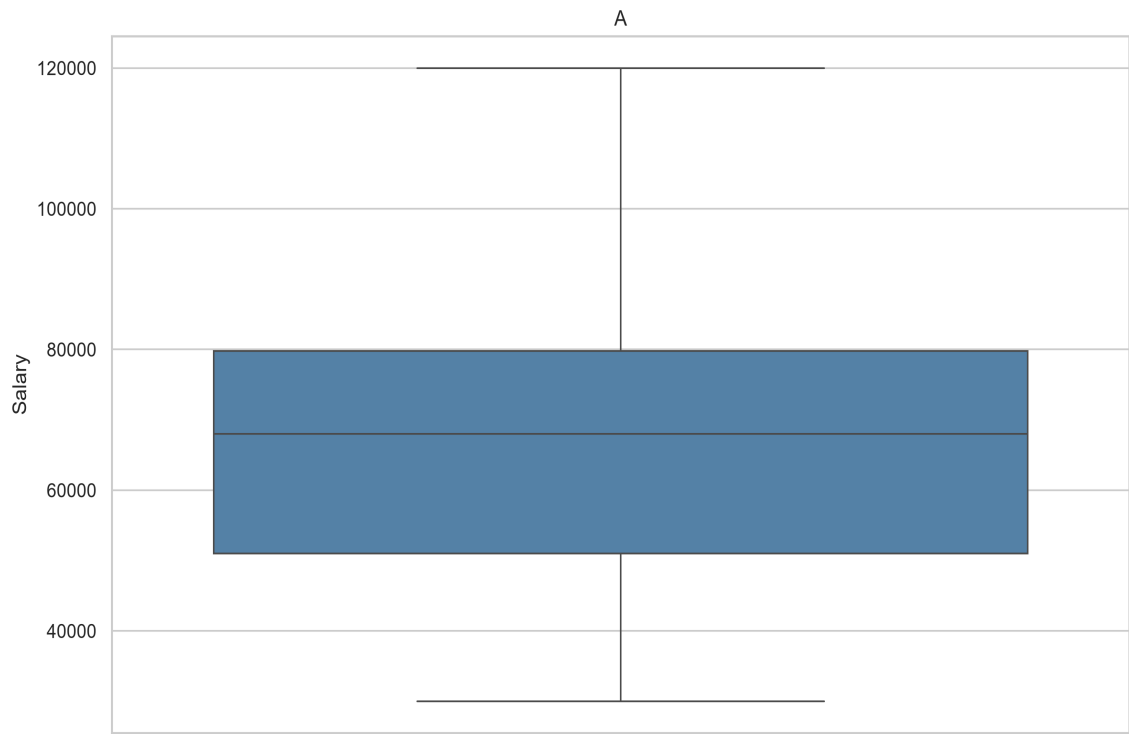
Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0

Salary	1
Performance	0
City	0

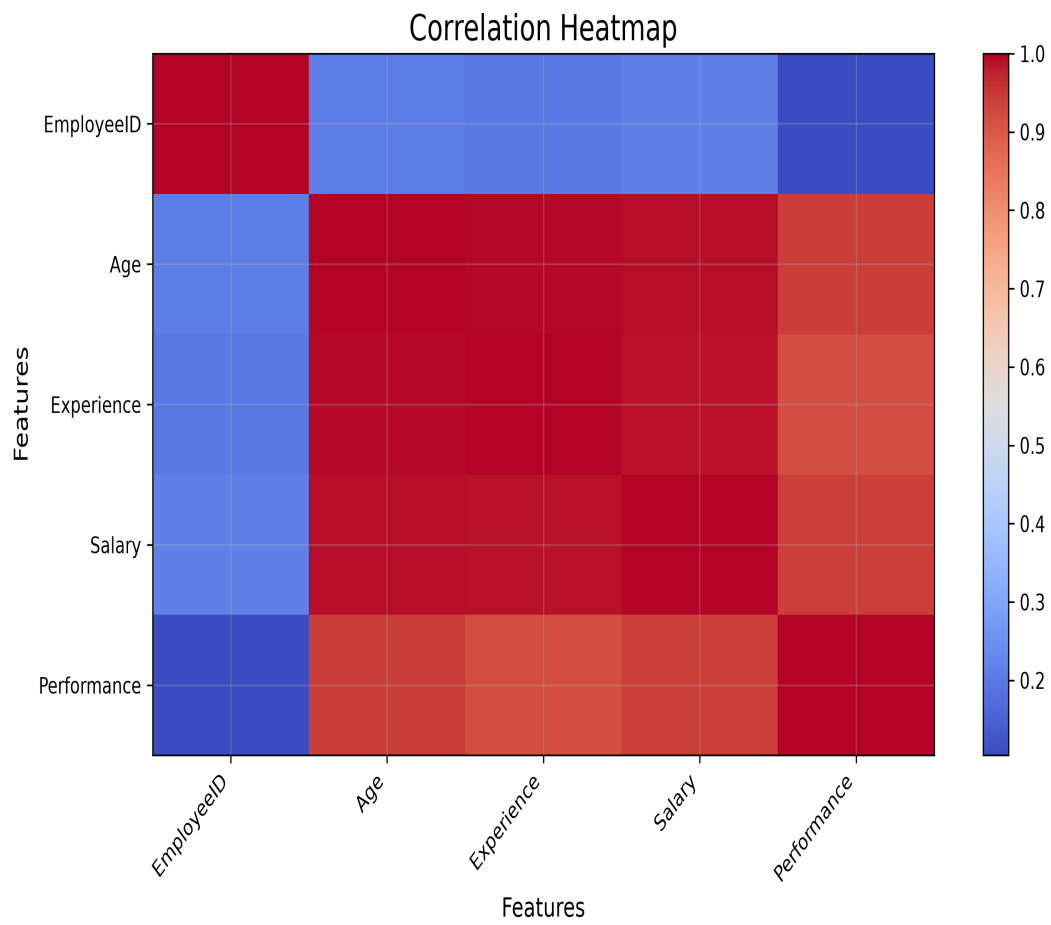
HISTOGRAM



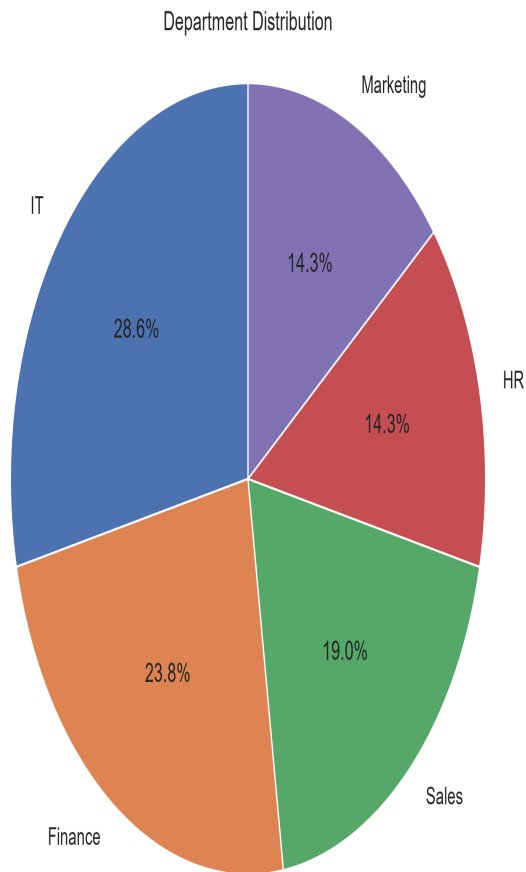
BOX PLOT



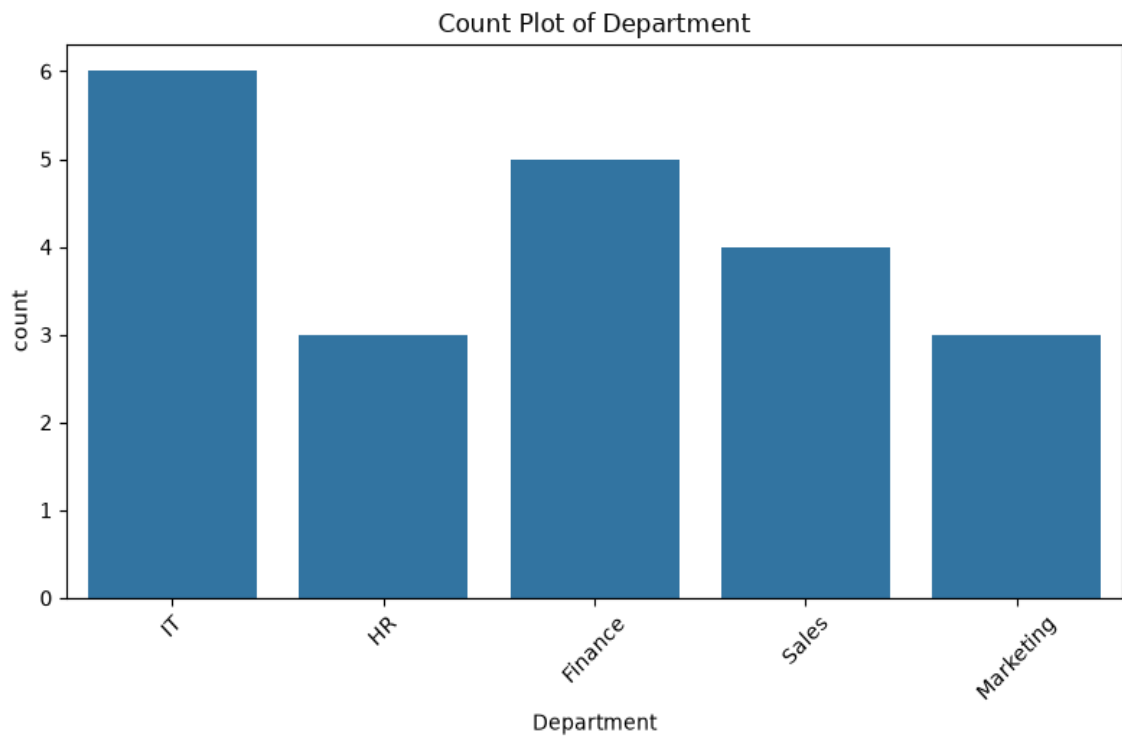
CORRELATION HEATMAP



PIE CHART



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

1. ****Perform Data Cleansing****:

* Remove the ****1 duplicate row**** immediately.

* Address the ****1 missing salary value**** using regression imputation, leveraging the 0.99 correlation with age and experience to input an highly accurate estimate.

* Investigate the ****1 experience outlier**** to ensure it is not a typo (e.g., entering "45" years of experience for a 35-year-old).

2. ****Audit the Performance Appraisal System****: Introduce objective, Key Performance Indicator (KPI)-based evaluations to ensure performance scores are decoupled from pure tenure.

3. ****Introduce Merit-Based Compensation Elements****: Supplement the seniority-based pay structure with spot bonuses, performance-based incentives, or fast-track promotions to retain high-potential junior employees.

4. ****Implement HR Analytics Dashboards****: Build the recommended visualizations (Correlation Heatmap, Scatter Plots, and Categorical Count Plots) to provide HR leadership with real-time visibility into compensation equity and demographic distributions.

Rule-based Recommendations

- ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ■■ 1 potential outliers detected.
- ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.